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SEC:B

SUBJECT:DBMS

PRACTICAL 3

Extracting data using patterns

Note: In sample code, change the table name and attributes according to your code.

```
SQL> create table cus(  
  2  id number primary key,  
  3  name char(10),  
  4  city char(10)  
  5  );
```

Table created.

```
SQL> insert into cus(id,name,city)  
  2  values(1,'alfreds futterkiste','berlin');  
values(1,'alfreds futterkiste','berlin')  
      *  
ERROR at line 2:  
ORA-12899: value too large for column "SYSTEM"."CUS"."NAME" (actual: 19,  
maximum: 10)
```

```
SQL> insert into cus(id,name,city)  
  2  values(1,'alfreds','berlin');
```

1 row created.

```
SQL> insert into cus(id,name,city)  
  2  values(1,'alfreds','berlin');  
insert into cus(id,name,city)  
      *  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C0011097) violated
```

```
SQL> insert into cus(id,name,city)  
  2  values(2,'antrujillo','mexico');
```

1 row created.

```
SQL> insert into cus(id,name,city)  
  2  values(3,'antoniomoreo','mexico.df');  
values(3,'antoniomoreo','mexico.df')  
      *  
ERROR at line 2:
```

```

SQL> insert into cus(id,name,city)
  2  values(2,'antrujillo','mexico');

1 row created.

SQL> insert into cus(id,name,city)
  2  values(3,'antoniomoreo','mexico.df');
values(3,'antoniomoreo','mexico.df')
      *
ERROR at line 2:
ORA-12899: value too large for column "SYSTEM"."CUS"."NAME" (actual: 12,
maximum: 10)

SQL> insert into cus(id,name,city)
  2  values(3,'antonio','mexico.df');

1 row created.

SQL> insert into cus(id,name,city)
  2  values(4,'horn','london');

1 row created.

SQL> insert into cus(id,name,city)
  2  values(5,'berglunds','lulea');

1 row created.

SQL> select*from cus;

      ID NAME          CITY
-----
1 alfreds    berlin
2 antrujillo  mexico
3 antonio    mexico.df
4 horn       london
5 berglunds  lulea

```

SQL LIKE Operator

The LIKE operator is used in a WHERE clause to search for a specified pattern in a column. There are two wildcards often used in conjunction with the LIKE operator:

- The percent sign % represents zero, one, or multiple characters
- The underscore sign _ represents one, single character

Example

Select all customers that starts with the letter "a":

```
SELECT * FROM Customers
```

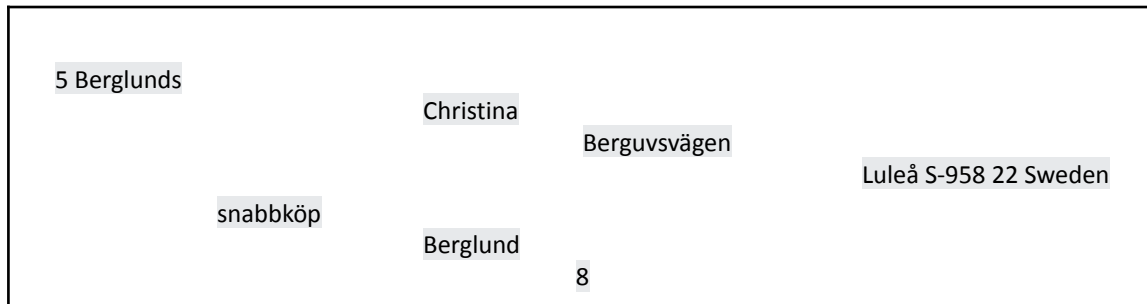
WHERE CustomerName LIKE 'a%';

```
SQL> select*from cus
2  where name like 'a%';
```

ID	NAME	CITY
1	alfreds	berlin
2	antrujillo	mexico
3	antonio	mexico.df

Demo Database

CustomerID	CustomerName	2) Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo	Ana Trujillo Avda. de la Constitución	México	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno Mataderos	México	05023	Mexico
4	Around the Horn	Thomas Hardy 120 Hanover Sq.	London	WA1 1DP	UK



The _ Wildcard

The _ wildcard represents a single character.

It can be any character or number, but each _ represents one, and only one, character.

Example

Return all customers from a city that starts with 'L' followed by one wildcard character, then 'nd' and then two wildcard characters:

```
SELECT * FROM Customers
WHERE city LIKE 'L_nd__';
```

```
SQL> select*from cus
  2  where city like 'L_nd__';

no rows selected
```

The % Wildcard

The % wildcard represents any number of characters, even zero characters.

Example

Return all customers from a city that *contains* the letter 'L':

```
SELECT * FROM Customers
WHERE city LIKE '%L%';
```

```
SQL> select*from cus
  2  where city like '%m%';

  ID NAME          CITY
-----
  2 antrujillo     mexico
  3 antonio       mexico.df
```

Starts With

To return records that starts with a specific letter or phrase, add the % at the end of the letter or phrase.

Example

Return all customers that starts with 'La':

```
SELECT * FROM Customers
```

```
WHERE CustomerName LIKE 'La%';
```

```
SQL> select*from cus
2  where name like 'be%';

-----
ID NAME          CITY
-----
5 berglunds      lulea
```

Tip: You can also combine any number of conditions using AND or OR operators.

Example

Return all customers that starts with 'a' or starts with 'b':

```
SELECT * FROM Customers
```

```
WHERE CustomerName LIKE 'a%' OR CustomerName LIKE 'b%';
```

```
SQL> select*from cus
2  where name like 'be%'or name like 'a%';

-----
ID NAME          CITY
-----
1 alfreds        berlin
2 antrujillo      mexico
3 antonio         mexico.df
5 berglunds       lulea
```

Ends With

To return records that ends with a specific letter or phrase, add the % at the beginning of the letter or phrase.

Example

Return all customers that ends with 'a':

```
SELECT * FROM Customers
```

```
WHERE CustomerName LIKE '%a';
```

Tip: You can also combine "starts with" and "ends with":

Example

Return all customers that starts with "b" and ends with "s":

```
SELECT * FROM Customers
```

```
WHERE CustomerName LIKE 'b%s';
```

```
SQL> select*from cus
  2  where name like '%a';

no rows selected

SQL> select*from cus
  2  where name like 'b%s';

no rows selected

SQL> select*from cus
  2  where name like 'b%a';

no rows selected
```

Contains

To return records that contains a specific letter or phrase, add the % both before and after the letter or phrase.

Example

Return all customers that contains the phrase 'or'

```
SELECT * FROM Customers
```

WHERE CustomerName LIKE '%or%';

```
SQL> select*from cus
2  where name like '%or%';
```

ID	NAME	CITY
4	horn	london

Combine Wildcards

Any wildcard, like % and _, can be used in combination with other wildcards.

Example

Return all customers that starts with "a" and are at least 3 characters in length: SELECT * FROM Customers

WHERE CustomerName LIKE 'a__%';

```
SQL> select*from cus
2  where name like 'a__%';
```

ID	NAME	CITY
1	alfreds	berlin
2	antrujillo	mexico
3	antonio	mexico.df

Example

Return all customers that have "r" in the second position:

SELECT * FROM Customers

WHERE CustomerName LIKE '_r%';

Without Wildcard

```
SQL> select*from cus
2  where name like '_r%';
```

If no wildcard is specified, the phrase has to have an exact match to return a result.

Example

Return all customers from Spain:

```
SELECT * FROM Customers
```

```
WHERE Country LIKE 'Spain';
```

```
SQL> select*from cus  
2  where city like'mexico';
```